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Emergy and emergy algebra explained by means of ingenuous set theory

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ABSTRACT

Emergy is an important concept that has originated several effects in ecology, systems ecology and sustainability science. Its communication, however, has always presented several problems, since it does not follow the same rules of conservation as other energy-based approaches. Attempts have been made to clarify emergy by means of more formal/mathematical approaches, but the problem persists. In this paper, we have introduced a view of emergy and of its algebra based on ingenuous set theory. By means of this simple tool, emergy can be defined as the set of solar exergy that is directly and indirectly necessary to make a product. The operation that correctly sums the emergy "carried" by the inputs to a process is the union. This definition and the operation of union are able to account for all the rules of emergy algebra.

1. Introduction

Emergy is a term deriving from the contraction of EMbodied enERGY; emergy evaluation is a thermodynamics-based approach that allows us to analyse production processes, ecosystems and nations on the basis of the past environmental work necessary to their formation and development. In this type of analysis, the various inputs that feed a given system are considered on the common basis of solar energy, that is the energy that underlies all the processes occurring in the biosphere. Emergy (indicated by *Em*) is defined as *the available solar energy used up directly and indirectly to make a service or product*. Its unit is the solar emjoule (abbreviated sej) (Odum, 1996).

Emergy is an extensive function, and its intensive correspondent is the (solar) *transformity* (abbreviated τ), which represents energy quality and is defined as *the solar energy required to make one joule of a service or product* (Odum, 1988). It was originally defined as the ratio of input emergy to output energy, but was subsequently revised as the ratio of input emergy to output exergy (Sciubba and Ulgiati, 2005; Bastianoni et al., 2007) to better take into account the different abilities to do actual work by different energy types.

The term *emergy* may also be considered as "energy memory", namely the memory of all the solar energy used up during a process; the emergy approach accounts for the work that the environment has carried out to make a service or product. Unlike energy, emergy is not a conservative function: its algebra follows a logic of memorization and not of conservation. The term "energy memory", as described by Scienceman, implies a "memory algebra" which does not obey the "energy conservation algebra": the emergy algebra was designed to give a quantitative account of non-conservative energy transformations (Scienceman, 1997).

Emergy is an important concept in ecology since it is able to consider all the processes that sustain the biosphere and the systems therein, on a common basis. Emergy is also used to evaluate the importance of nature's inputs to human economy (Odum, 1988).

Emergy is also one of the functions that plays the role of orientor (Müller and Leupelt, 1998), since Odum used it to modify Lotka's maximum power principle to encompass the distinctive role of different types of energy. For a complete overview of this argument please refer to Fath et al. (2001) and Jørgensen et al. (2007).

The main characteristics that distinguish emergy evaluation regard processes that simultaneously produce multiple outputs that are divided into two categories: *splits* and *co-products*. We have *co-products* if a process that produces two or more outputs cannot produce one without producing the other(s); on the other hand, if it is possible to arbitrarily separate outflows, we have *splits*. In other words, the difference between splits and co-products is the degree of freedom that we have in allocating energy (or mass) to the different products: if multiple outputs are obtained in fixed proportions (no degree of freedom), then we have co-products; if the energy in each of the outputs can be any amount, provided that the total quantity is constant (total degree of freedom), then the products are splits.

According to Brown, *emergy algebra* can be described with four rules (Brown and Herendeen, 1996):

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Rule 1. All source emergy of a process is assigned to process output.
Rule 2. Co-products of a process have the total emergy assigned to each pathway (no allocation).
Rule 3. When a pathway splits, emergy is assigned to each 'leg' of the split according to the percentage of total energy flow on the pathway.
Rule 4. Emergy cannot be counted twice within a system: (a) emergy in feedback cannot be double counted; (b) co-products, when reunited, cannot add up to a sum greater than the source emergy from which they were derived.

It follows that the transformities of two (or more) splits are identical, while their emergy content is generally different (unless energy is equally distributed among the legs); on the contrary, the transformities of co-products are generally different (unless the energies of the various outputs are equal).

With all the precautions found in Rule 4, Emergy is often summarized with the formula (see for example Sciubba and Ulgiati, 2005)

$$Em = \sum_i \tau_i Ex_i \tag{1}$$

where Em is the emergy, τ_i is the transformity and Ex_i is the exergy of the *i-th* input.

However, no emergy analyst should apply this equation carelessly, for it does not consider Rule 4 and the importance of distinguishing between splits and co-products.

2. Emergy as a set

Several attempts have already been made to mathematically illustrate emergy and its properties in a more structured way (e.g. Giannantoni, 2006; Giannantoni et al., 2005; Li et al., 2010). Some of these are difficult to translate into a more practical form, while others are not fully compliant with emergy theory; we have sought a more friendly way to illustrate emergy and its rules through using sets and ingenuous set theory.

Here we define emergy as the *set* of the solar exergy that is directly and indirectly required to obtain a product; since a product (or service) is obtained through the convergence of several inputs, its emergy can be seen as the *union* of the sets of solar exergy required to obtain each input. Therefore, we can define emergy as:

$$Em = \bigcup_{i=1}^{n} \{\text{direct and indirect solar exergies}\} \tag{2}$$

where n is the total number of inputs.

As a consequence, $Em = \bigcup_i \tau_i Ex_i$ which corrects Eq. (1) and, as later shown, it is able to fully consider the rules of emergy algebra.

To be more specific, we can say that emergy is the set of all the (equivalent) photons falling on certain portions of the biosphere in certain time intervals, that have been used directly and indirectly to make a product or service. The word "equivalent" is necessary in order to include the two main energies not coming from solar radiation, i.e. tides and deep geological heat (Odum, 1996).

2.1. Emergy and set theory

We shall now give a more formal description of emergy by means of sets and ingenuous set theory.[1] Let S be a system, and let D_S be its associated diagram. D_S is made up of the following parts:

1. *Arrows*: They represent physical flows (quantifiable in exergy terms), that can be translated into emergy.
2. *Blocks*: Every block represents a process of S, i.e. *the operation on flows*.

In particular, since arrows connect the blocks, we will distinguish between the arrows that are incoming to a block, and the arrows that are outgoing from a block. A process transforms the two or more exergy flows into one (or more) exergy output(s). In this process the emergy of the inputs is further memorized and the output has an emergy that is the union of the emergy inputs.

If an arrow i connects a block B_2 to a block B_1, and B_1 precedes B_2 in the intuitive order of transformations (from left to right) in D_S, then we say that i is a feedback. This is consistent with emergy literature and the way Odum describes the use of energy system diagrams: "In the diagram energy is successively transformed in passing from left to right with available energy being degraded... Pathways from right to left are high quality flows that feed back to the left to interact and amplify the inflows of larger quantity" (Odum, 1991).

Subsequently, we shall provide an interpretation of arrows and blocks in the framework of (ingenuous) set theory. In particular, we will interpret each arrow as the set of (direct and indirect) photons that reaches an area, in a certain time interval, by means of a given physical flow, while each block will be interpreted by means of ad-hoc defined set-theoretical operations.

2.2. Systems without feedback

Let S be a system with no feedback. In this case, we can describe S by means of the following model: first of all, we will identify every arrow i of D_S by a set that we will denote as X_i. It is convenient to consider X_i as the set of all direct and indirect photons that fell down in a given portion of the biosphere at a given time (emergy). X_i thus gives a representation of the emergy that is embodied in that flow. In other words, we are interpreting the amount of emergy (associated with a physical input) by the set X_i.

Every arrow *per se* is invariant with respect to space and time; different arrows, i.e. different flows (and hence different emergies) are distinguished whenever the photons in X_i fell down in different portions of the biosphere and/or at different times. Therefore, it is convenient to consider every set X_i as if it was labeled by two parameters (which we will not explicitly write): one for space (representing where the photons fell down), and one for time (stating when the photons fell down).

Before entering into details, let us spend some words to give the basic ideas and intuitions:

Inputs. Any input is interpreted by means of a set X (its emergy), that is regarded both as its cardinality (in this way we describe the total amount of emergy "entering" the system via the input X, and so we provide a quantitative interpretation of emergy flows) and as the collection of its elements (including their spatial and temporal characteristics, and in this way we give a further qualitative interpretation of emergy flows within S). The latter interpretation (the qualitative one) is crucial to understanding why our model is capable of satisfying the basic principle of emergy algebra, i.e. that emergy cannot be counted twice within a system. This is due to the fact that two sets X and X', despite being quantitatively similar, can be interpreted as radically different entities because of their

[1] The ingenuous set theory considers sets as collections of objects (elements) and it is different from axiomatic set theory because it considers sets only those that satisfy certain axioms. Generally, the term "set theory" refers to the axiomatic theory but if we want to use set theory as a mere instrument to be applied in various fields, we can refer to the ingenuous set theory.

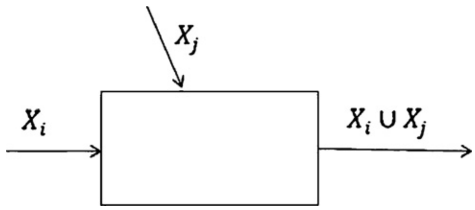


Fig. 1. Example of a process with two inputs.

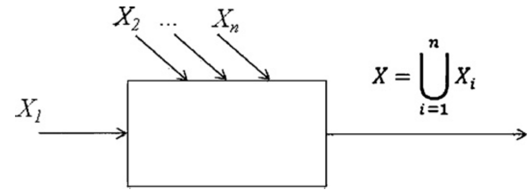


Fig. 2. Example of a process with *n* inputs.

qualitative interpretation, recalling that every set in our model also brings spatio-temporal information.

We shall now clarify this situation with some examples.

Let us assume that two inputs supply a system S through the arrows i and j interacting in a given block of S (see Fig. 1). The convergence of emergy in a block of S is an operation that is modeled by taking the union of X_i and X_j. Therefore, the total emergy of the output of the block in S, is quantified by the well-known formula:

$$\left|X_i \cup X_j\right| = \left|X_i\right| + \left|X_j\right| - \left|X_i \cap X_j\right| \tag{3}$$

where the symbol $|\,|$ represents the cardinality of the set.

In this way we easily avoid the double-counting. On one hand, if we assume that X_i and X_j represent photons that have either converged to S from different (and eventually far) areas, or in different temporal periods, we then interpret X_i and X_j as two disjoint sets (i.e. $X_i \cap X_j = \varnothing$). In this case we compute:

$$\left|X_i \cup X_j\right| = \left|X_i\right| + \left|X_j\right| - \left|X_i \cup X_j\right| = \left|X_i\right| + \left|X_j\right| \tag{4}$$

On the other hand, let us assume that X_i has spatio-temporal characteristics that we also find in X_j and that set X_i has cardinality strictly less than the cardinality of X_j, that is $|X_i| < |X_j|$. This means that X_i is a collection of photons that already appeared in X_j, thus X_i and X_j have, at least partially, originated as co-products, and we can avoid the double-counting effect. In other words, X_i is a strict subset of X_j (i.e. $X_i \cap X_j = X_i$), so we compute:

$$|X_i \cup X_j| = |X_i| + |X_j| - |X_i \cap X_j| = |X_i| + |X_j| - |X_i| = |X_j| \tag{5}$$

hence our model avoids double counting.

Finally, we would like to stress that this distinction would be impossible if we only looked at a set as its cardinality, and we would lose the further essential information brought by the qualitative interpretation (spatio-temporal labels).

Outputs. Outputs, as well as inputs, are represented by sets. As we briefly discussed above, the outputs of a system S can either have the form of a split, a co-product, or we may otherwise call them simple outputs. Since simple outputs do not need to be investigated further, we shall now discuss splits and co-products. In the following we assume, without loss of generality, that splits and co-products consist of just two outgoing arrows. If i is the main output from which two splits are obtained, then $\pi_1(X_i)$ and $\pi_2(X_i)$ provide a partition of X_i. The "*partition*" of a set X is a division of X into non-overlapping and non-empty parts that cover all of X." (Brualdi, 2004). In particular this means that $\pi_1(X_i) \cap \pi_2(X_i) = \varnothing$ and $\pi_1(X_i) \cup \pi_2(X_i) = X_i$.

This interpretation is coherent with the fact that the two outputs of a split must sum up to the set outgoing from the process (X_i). Therefore using same Eq. (3) we compute

$$\left|\pi_1(X_i) \cup \pi_2(X_i)\right| = \left|\pi_1(X_i)\right| + \left|\pi_2(X_i)\right| - \left|\pi_1(X_i) \cap \pi_2(X_i)\right| = \left|X_i\right| \tag{6}$$

as required.

If i' and i'' are two co-products, then we require that $X_{i'} = X_{i''}$. In this case we have

$$\left|X_{i'} \cup X_{i''}\right| = \left|X_{i'} \cap X_{i''}\right| = \left|X_{i'}\right| = \left|X_{i''}\right| \tag{7}$$

Notice that, by this requirement, any co-products are described, within our model, by the same set.[2] This interpretation is consistent with what we stated before about the multiple outputs as co-products.

Blocks: It is convenient to distinguish two stages in the process represented by the block we are considering. The first one, which is the same in every block of D_s, consists in collecting all the inputs to the block (i.e. the sets represented by the entering arrows), by taking as operation the union of all these sets. The second stage consists in analyzing the output process: from the union defined in the previous stage, the block either produces a simple arrow, splits, or co-products.

In particular, let X be the set resulting from the union of all the inputs-sets. Then:

(a) In the case of a split of X, the model produces a *partition* of X into two[3] subsets that we denote by $\pi_1(X)$ and $\pi_2(X)$.
(b) In the case of two co-products, since emergy does not change from the total emergy represented by X, to each of the two branches of a co-product, we assign a copy of X.

Since the system has no feedback, no other cases are needed to describe the model.

2.3. Systems with feedback

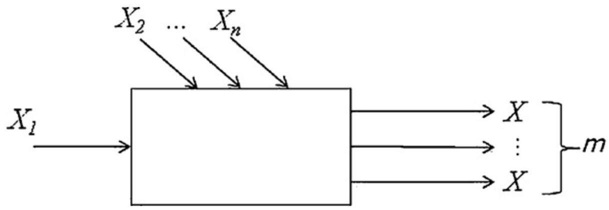
Let us now consider a system that has at least one feedback input. As it is clear from what we have stated above, every set representing emergy has to be considered with its spatio-temporal labels. These labels and their updating will play a special role in this section. We shall now distinguish the following cases:

Split feedback. Consider a split of a set X and assume that one path of the split feeds back. Let us assume, without loss of generality, that $\pi_1(X)$ is the output that feeds back. In order to calculate the set representing the amount of emergy after the feedback is added, we proceed as follows: if we consider a flow $\pi_1(X)$ feeding back from one block to another or to the same block, a problem arises in dealing with the time in which the output splits and the time in which the feedback is utilized as input. First the split feeding back is collected and then it is used in the next cycle.[4] The correct representation of this situation would be with at least a time delay along the arrow feeding back. For this reason we have to transform $\pi_1(X)$ into its *disjoint copy* $\pi_1(X)'$ that only has the time characteristics

[2] Strictly speaking, any co-product produces a multiset (see Blizard, 1989). It is not the aim of this paper to stress the distinction between multiset theory and ingenuous set theory. Here we have used concepts that are quite intuitive and that do not require a more formal treatment.

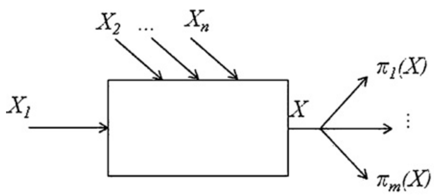
[3] We are assuming that splits and co-products have just two outputs, without any loss of generality; the general case in which splits and co-products have more than two outputs follows the same reasoning.

[4] Imagine the case of a corn field: the grains collected in one cycle are then used in the next one. The same holds for shorter cycles: natural cycles are almost always annual, while more human-oriented cycles can have any periodicity, including very short ones in the order of magnitude of seconds.



$$X = \bigcup_{i=1}^{n} X_i$$

Fig. 3. Example of a process with n inputs and m co-products (with $m \geq 2$ and generally, $m \neq n$).



$$X = \bigcup_{i=1}^{n} X_i \qquad \bigcup_{j=1}^{m} \pi_j(X) = X \qquad \pi_j(X) \cap \pi_k(X) = \emptyset \text{ con } j \neq k$$

Fig. 4. Example of a process with n inputs and m splits (with $m \geq 2$ and generally, $m \neq n$).

different from $\pi_1(X)$; and therefore we have to treat $\pi_1(X)'$ as any other independent input.

Co-product feedback. The situation in which the feedback has originated from a co-production is slightly easier. In fact, in this case, we need not consider a disjoint copy of the output we are feeding back, but rather a copy X without further modifications. This is essentially due to the fact that, in a co-product feedback, the spatio-temporal labels we attach to a set lose all meaning, because all the outputs of the coproduction bring the same emergy, independent of space and time: while in the case of splits the feedback requires further emergy with respect to the net output, in the case of co-products the feedback has not required any "fresh" emergy, since it was obtained together with another product.

3. Emergy algebra interpreted with ingenuous set theory

Let us now verify that the definition of emergy with sets and the use of union as the fundamental operation are compliant with the rules of emergy algebra.

Rule 1. All source emergy of a process is assigned to process output (see Fig. 2).

It is evident that the union has an advantage with respect to the sum operation, since it does not require any special restriction: if the emergy flows are independent then we have to sum them, otherwise only the independent parts (without intersections) should be added.

Rule 2. Co-products of a process have the total emergy assigned to each pathway (no allocation) (see Fig. 3).

Co-production is translated into an m-replication of the emergy set (where m is the number of co-products). This has implications in the results of an interaction of co-products in a cascade of processes or in feedbacks (see Rule 4).

Rule 3. When a pathway splits, emergy is assigned to each 'leg' of the split according to the percentage of total energy flow on the pathway (see Fig. 4).

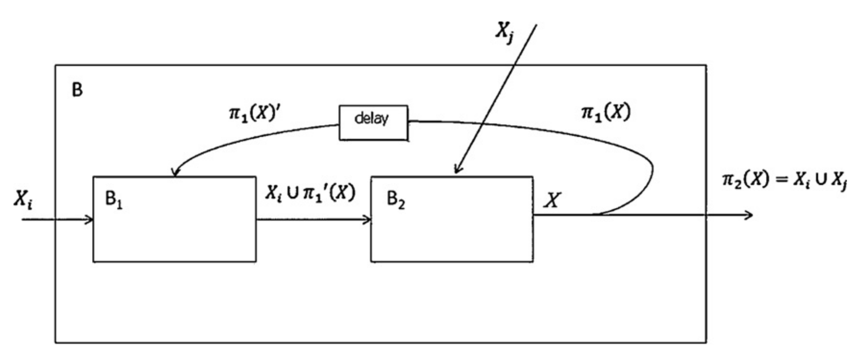
The emergy of each split is a portion (element of the partition) of the total emergy of the main process. If the splits interact after the process they should be added (the union is of independent emergies). For the case in which one of the splits feeds back, see Rule 4.

Rule 4. Emergy cannot be counted twice within a system:

(a) emergy in feedback cannot be double counted;

If the feedback is a split, by definition the transformity of all the outflows (entire flow and splits) has to be identical. Brown and Herendeen (1996) interpreted this fact first assigning the emergy of the main outflow and then distributing this emergy to the splits, according to their exergy content. The set approach, instead, shows that the feedback has to be considered in the evaluation, since the emergy of the feedback does not have the same time characteristics as the other flows: it has to be produced in a previous production cycle to be available in the present time and stored somewhere within the system. It is therefore an independent input, and when we have to operate the union of the sets the result is the addition of the split feeding back (see Fig. 5).

The time problem is easily understandable for systems in which a "natural" cycle occurs (yearly, circadian, etc.): in an



$$X = X_i \cup X_j \cup \pi_1{}'(X)$$

Fig. 5. Example of a process with a split that feeds back. The small box along the feedback means that there is a delay between the time in which the feedback is withdrawn and the time it is actually used as input. The union of the outputs that exit the larger box (B) must be equal to the union of inputs to B. In this way, the value of X is forced and the transformities of the output of the smaller box (B_2) and of the splits is identical.

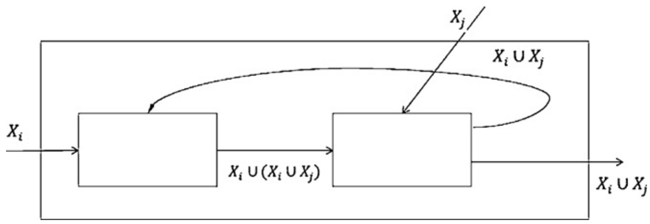


Fig. 6. Example of a process with a co-product that feeds back: only the independent part of the emergy of the feedback is considered.

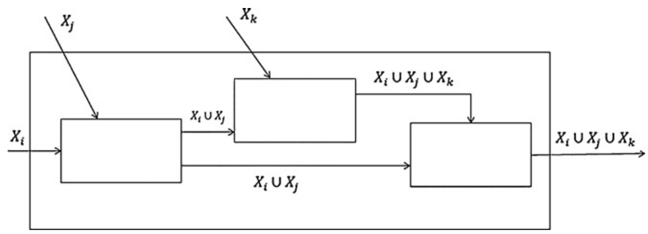


Fig. 7. Example of a process where co-products are reunited: we have no double counting.

agricultural field the seeds collected in one year will be used the following one. This is less evident in a continuous system where the dynamics are much faster: in an electricity power plant, part of the production is used for self-sustaining the system (control mechanisms, lighting, etc.) but the considerations can be exactly the same if we consider a time frame of the order of seconds.

If the feedback is a co-product, the emergy of the feedback is the total emergy of the output. When the flow interacts back, the operation of union of these sets implies that only the independent part of the emergy of the co-product feedback should be considered (see Fig. 6).

(b) co-products, when reunited, cannot add up to a sum greater than the source emergy from which they were derived.

The operation of union makes it impossible to have a sum greater than the source emergy; if co-products reunite, the emergy of the co-product is counted only once if all co-products are interacting, or the maximum is considered if one or more co-products interact only partially (see Fig. 7). This is always valid since, as we highlighted earlier, in co-production the importance of time characteristics is lost.

4. Conclusions

Emergy can be represented as a set, and we can use ingenuous set theory to elaborate on this concept. One of the peculiarities of emergy theory, i.e. the existence of two different types of multiple outputs (splits and co-products), is easily addressed by associating the operation of partition to the splits and the operation of copy to co-products. Every process in which one or more inputs (each with its emergy) converge to produce an output can be represented with the operation of union. In this way the necessity of an emergy algebra is somehow reduced, since all its rules can be encompassed in the property of union of sets: co-products and feedback are never counted twice when they interact in a process.

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